

PART A

LITERATURE REVIEW 1

Title	Customer Churn Prediction in Banking—Machine Learning Application
Harvard reference	Hoang, H.T.T., Dang, T.V. and Do, L.T.D. (2026) 'Customer Churn Prediction in Banking—Machine Learning Application', Applied Computational Intelligence and Soft Computing, 2026, Article 8889093. doi: 10.1155/acis/8889093.
Link	https://onlinelibrary.wiley.com/doi/10.1155/acis/8889093
Modelling	Seven supervised classification models—Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM), K-Nearest Neighbours (KNN), LightGBM, and XGBoost—are compared in this work. GridSearchCV with stratified 5-fold cross-validation was used to tune the hyperparameters, with AUC-ROC serving as the optimisation criterion. Tree-based and ensemble methods are offered by Random Forest, XGBoost, and LightGBM. L1 and L2 regularisation are also used by XGBoost. The study also uses a cost matrix to provide a business-oriented classification threshold, K-means clustering to generate customer risk categories, and permutation feature importance to find significant predictors.
Preprocessing techniques	RowNumber, CustomerID, Surname, and Point Earned were among the identifying factors that the researchers eliminated. Due to its substantial correlation with the result, Complain was removed since it was deemed data leakage. Unordered categories were one-hot encoded, binary variables were binary encoded, and categorical variables were numerically encoded. AgeGroup, Engaged, Tenure_Age_Ratio,

	Balance_Salary_Ratio, and ZeroBalance are the five engineering variables that were developed. For models that were sensitive to distance, continuous variables were standardised. To solve class imbalance, the dataset was divided before applying SMOTE + ENN to the training data; the test set kept its original distribution.
Datasets used	About 10,000 customer records together with financial, behavioural, and demographic data are included in the public Kaggle Bank Customer Churn dataset. Whether a customer churned is indicated by the goal variable, Exited. There are about 20% churners and 80% non-churners in the dataset. The dataset availability information is made clear by the authors.
Key takeaways	1. Prevent data leakage: Use only variables that were available before to the prediction point. This is especially crucial for STADIOEquities since the model needs to forecast future dormancy or activation failure instead of using data generated after the event. 2. Address class imbalance: Techniques like SMOTE should only be applied to the training data since STADIOEquities' dormant/failed-activation clients might be a minority class. 3. Model interpretation is important: The STADIOEquities question of which observable client behaviours are most predictive and which clients should receive intervention can be addressed with the use of permutation feature importance and risk segmentation.

LITERATURE REVIEW 2

Title	Propension to customer churn in a financial institution: a machine learning approach.
Harvard reference	Lemos, R.A.L., Silva, T.C. and Tabak, B.M. (2022) 'Propension to customer churn in a financial institution: a machine learning approach', <i>Neural Computing and Applications</i> , 34, pp. 11751–11768. doi: 10.1007/s00521-022-07067-x.
Link	https://pmc.ncbi.nlm.nih.gov/articles/PMC8898559/?utm
Modelling	The Decision Tree, K-Nearest Neighbours, Logistic Regression, Elastic Net, Support Vector Machine with a radial kernel, and Random Forest are the six classification techniques that the authors compare. L1 (lasso) and L2 (ridge) regularisation are combined in Elastic Net. For model selection, the researchers employ a 90% training/10% test split and run 10-fold cross-validation ten times. The optimisation metric is AUC-ROC. Additionally, they combine the six classifiers into an ensemble and use a voting system to improve the weights of the component models.
Preprocessing techniques	Using summary functions like percentages and differences over six-month periods, the researchers developed behavioural characteristics. Closing accounts or ceasing account activity for a period of six months was their definition of churn. K-Nearest Neighbour imputation with $k = 3$ was used to address missing observations. Segment_FX, Accreditation, Portability_Request, Complaint_Request, Automatic_Debt_DIFF, Salary_Credit_DIFF, and Insurance_DIFF were among the variables with nearly zero variance that were eliminated. The numerical variables were then standardised.

Datasets used	A confidential customer-level dataset from a major bank in Brazil. A balanced sample of 500,000 clients was chosen at random from a population of over 9.7 million clients by the researchers. Financial features and information about client activity and relationships are included in the data. The first six months of the study are used to build predictors, and the next six months are used to determine whether the consumer churned.
Key takeaways	1. Make advantage of a temporal prediction structure: The study forecasts churn for the next six months using past data from one six-month period. This is particularly relevant to the suggested six-month dormancy prediction window of STADIOEquities. 2. Engineer behavioural features: Variations, percentages, and shifts in activity are more effective than static variables alone at capturing declines in consumer engagement. Similar functionalities can be created using STADIOEquities using deposits, trades, logins, product activity, and marketing interactions. 3. Rather than assuming that one method is the best, compare other models: The authors use repeated cross-validation to benchmark several classifiers, which offers a more robust modelling process than choosing a single algorithm without comparison.

LITERATURE REVIEW 3

Title	Prediction of bank credit customers churn based on machine learning and interpretability analysis
Harvard reference	Li, Y. and Yan, K. (2025) 'Prediction of bank credit customers churn based on machine learning and interpretability analysis', <i>Data Science in Finance and Economics</i> , 5(1), pp. 19–34. doi: 10.3934/DSFE.2025002.
Link	https://www.aimspress.com/article/doi/10.3934/DSFE.2025002
Modelling	Six machine-learning models—Random Forest (RF), Gradient Boosting Decision Tree (GBDT), Extra Trees, AdaBoost, XGBoost, and CatBoost—are compared by the researchers. They create several model/preprocessing combinations by training the models using four distinct class-balancing techniques: Random Oversampling, SMOTE, Borderline-SMOTE, and ADASYN. SHAP (SHapley Additive exPlanations) is then used to interpret the top-performing model. In order to look into the connections between factors and churn, the study also uses an R-learner causal inference method.
Preprocessing techniques	Because CLIENTNUM is an identification with no predictive value, the researchers eliminated it. For Naive Bayes modelling, two more variables from the original dataset were also eliminated. Numerical representations of six category variables were created. Dummy variables were created from the unordered categorical variable Marital_Status. The majority class of each variable was substituted for unknown values in Education Level, Marital Status, and Income Category. An 80:20 split was used to divide the data into training and

	test sets. Next, Random Oversampling, SMOTE, Borderline-SMOTE, or ADASYN were used to balance the training data.
Datasets used	The Kaggle Bank Customer Churn dataset, which has 10,127 customer observations and 23 variables, is used by the researchers. Personal traits, credit card details, and customer usage patterns are all included in the dataset. Attrition_Flag, the target variable, indicates if the customer is a current or attrited customer.
Key takeaways	1. Examine various class-balancing strategies: The study shows that techniques like SMOTE, Borderline-SMOTE, ADASYN, and Random Oversampling can be used to handle imbalanced customer-outcome data. 2. Combine explainability with prediction: SHAP can determine which factors have the biggest impact on churn forecasts. This clearly answers the study question posed by STADIOEquities regarding the characteristics and behaviours of clients that are most indicative of dormancy or activation failure. 3. Transaction behaviour can be very instructive: According to the study, key indicators linked to churn include transaction count, transaction amount, number of items, and changes in transaction activity. This facilitates examining the funding, trading, and behavioural aspects of STADIOEquities.

Publicly available dataset that will be used for PART B.

Name	Bank Customer Churn Dataset
Harvard reference	Hoang, H.T.T., Dang, T.V. and Do, L.T.D. (2026) 'Customer Churn Prediction in Banking–Machine Learning Application', Applied Computational Intelligence and Soft Computing, 2026, Article 8889093. doi: 10.1155/acis/8889093.
Link	https://www.kaggle.com/datasets/radheshyamkollipara/bank-customer-churn/code
Number of instances	10,000 customer records. There is no mention of distinct predetermined training and testing divides in the dataset. During the modelling process, the 2026 publication establishes its own modelling divide.
Number of feature columns	There were a total of 14 original columns/attributes. These consist of the target variable and the customer variables. RowNumber, CustomerId, Surname, CreditScore, Geography, Gender, Age, Tenure, Balance, NumOfProducts, HasCrCard, IsActiveMember, EstimatedSalary, and Exited were the original columns. Thus, there are 13 original predictor columns plus 1 target column if your assignment uses "feature columns" to mean predictor variables excluding the target.
Number of classes (if applicable)	Since this is a binary classification problem, there are two classes. There are two possible classifications for the target Exited: 0 = retained/not churned and 1 = exited/churned. According to the 2026 publication, the dataset includes roughly 7,962 retained consumers and 2,038 churned customers.

Target range (if applicable)	Since this is a classification problem rather than a regression problem, it is not appropriate. Exited {0, 1} is the binary/categorical target.
Origin	The public Kaggle Bank Customer Churn dataset is the source of the dataset. According to the journal, it includes 10,000 customer records from a commercial banking setting that include financial, behavioural, and demographic data. The information is used to forecast if a consumer has left or churned. Kaggle makes the data publicly accessible.